Process intensification of catalytic-wall Taylor-Couette reactors by reinforcement learning

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Abstract

López-Guajardo et al. [4] showed that pulsating the angular speed of the inner cylinder of a catalytic-wall Taylor-Couette reactor increases conversion relative to constant rotation at the same nominal speed, delivering process intensification without new hardware. We extend that idea from fixed, hand-designed waveforms to *state-dependent* modulation discovered by reinforcement learning (RL). A resolved-film CFD model of the reactor (OpenFOAM, Schmidt number $Sc = 1075$, $25\,\mu$m wall cells) is wrapped in a Gymnasium environment in which a TD3 agent re-parameterizes the rotation waveform every 10 s under a fixed-mean-speed constraint, making every policy an equal-power competitor by construction. The trained agent beats the best static waveform found by grid search on the training geometry ($R = X - P/P_{\max}$ of 0.289 vs. 0.275; conversion 0.406 vs. 0.392 at 3.7 W), using an interpretable schedule: spin-up, fast "film-renewal shimmy", deep slow pulses, and a terminal constant-speed harvest. Deployed zero-shot on the full-height reactor ($\Gamma = 30$, five times the residence time of the training geometry), the same policy reaches a final-window conversion of 0.902, beating both constant rotation (0.787) and the best static pulse (0.876) at equal power. We also show that part of the learned advantage is a finite-horizon "batch" effect, quantify the sustainable component, and discuss what changes are needed to optimize continuous steady-state operation.

1 Introduction

Process intensification usually means replacing equipment; modulating the operation of equipment already in place is far cheaper. Taylor-Couette (TC) reactors — the annular gap between a rotating inner cylinder and a fixed outer cylinder — offer predictable, tunable vortex flow and are used industrially in crystallization, electrolysis, and catalysis. When the outer wall is catalytic, overall conversion is limited by mass transfer of reactant across the concentration boundary layer at that wall. López-Guajardo et al. [4] demonstrated, in simulation, that *unconventional modulation* of the inner-cylinder speed — square-wave pulsation about a nominal speed w_b — thins and renews this boundary layer and can raise conversion at essentially the same (or lower) motor energy than constant rotation.

Fixed waveforms are only a small corner of the modulation design space: the duty cycle, period, and depth of pulsation could in principle vary in time and respond to the state of the flow. Deep reinforcement learning has recently proven able to discover non-obvious control strategies in fluid systems, including turbulent drag reduction [3] and enhanced heat transfer in convection [7]. Here we ask the analogous question for catalytic TC reactors: **can state-dependent, learned modulation beat the best static waveform at equal power?**

Our contributions are:

1. a resolved-film CFD benchmark of the pulsed-vs-constant effect of Ref. [4] on both a short ($\Gamma = 6$) and the full-height ($\Gamma = 30$) reactor geometry — a grid over the full waveform family on the conversion-vs-power plane — confirming that pulsation beats constant rotation, that the advantage grows substantially in the tall reactor, and identifying a static champion waveform substantially stronger than the paper's own configuration;

2. a Gymnasium [5] RL environment in which a TD3 agent [2] chooses block-wise waveform parameters under a fixed-mean-speed (equal-power) constraint, together with the training fixes (terminal bootstrapping, high update-to-data ratio) required to make TD3 converge on a few thousand expensive CFD transitions;

3. a learned policy that beats the static champion on the training geometry and transfers zero-shot to the full-height reactor, where it outperforms every static waveform tested; and

4. an analysis separating the sustainable part of the learned gain from a finite-horizon "harvest" effect, with implications for applying episodic RL to continuously-operated process equipment.

2 Methodology

2.1 Reactor model

For the sake of knowing whether our RL approach outperforms the static waveforms of López-Guajardo et al. [4], our formulation closely follows theirs. The reactor geometry is summarized in Table 1. Two variants are used: a *short* reactor ($H = 38.1$ mm, $\Gamma = 6$), one fifth of the paper's height, which keeps RL training affordable; and the *full-height* reactor of the paper ($H = 190.5$ mm, $\Gamma = 30$), used as a transfer target. The computational domain is a $5°$ azimuthal wedge of the annulus. Fresh reactant enters through the top annular face and leaves through a one-cell side-outlet band at the bottom of the outer wall; at a feed rate of 100 mL/min the mean residence time is $\tau \approx 26$ s in the short reactor and ≈ 130 s in the full-height one.

Table 1: Geometry of the catalytic-wall Taylor-Couette reactor. The computational domain is a 5° azimuthal wedge of the full annulus, with a one-cell side-outlet band at the bottom of the outer wall.

Parameter	Symbol	Value
Inner cylinder radius	R_i	25.40 mm
Outer cylinder radius	R_o	31.75 mm
Gap width	$d = R_o - R_i$	6.35 mm
Reactor height (short / full)	H	38.10 / 190.5 mm
Radius ratio	$\eta = R_i/R_o$	0.80
Aspect ratio (short / full)	$\Gamma = H/d$	6.0 / 30
Side-outlet band height	h_{out}	0.42 mm
Azimuthal wedge angle	θ	5°

The incompressible Navier-Stokes equations

$$\nabla \cdot \mathbf{u} = 0, \tag{1}$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u}, \tag{2}$$

and the transport equation for the dilute reactant concentration c

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = D \nabla^2 c, \tag{3}$$

were solved by the finite-volume method in OpenFOAM v2506 [6] (transient `pimpleFoam` with a passive scalar, adaptive time stepping at a Courant limit of 1.8). The working fluid is a silicone-oil-like liquid ($\nu = 1.075 \times 10^{-5}$ m^2/s, $\rho = 930$ kg/m^3) carrying a reactant with molecular diffusivity $D = 1 \times 10^{-8}$ m^2/s, giving a Schmidt number $Sc = \nu/D = 1075$. The feed concentration is $c_0 = 50$ mmol/m^3.

The catalytic outer wall is modeled as a perfect sink, $c = 0$, so consumption is limited by molecular diffusion across the concentration boundary layer — the mass-transfer-limited regime [1]. At $Sc \sim 10^3$ this layer is far thinner than the momentum boundary layer, so the mesh is graded radially to a 25 μm first cell at the outer wall (3,330 cells for the short reactor, 16,650 for the full-height one), resolving the film *directly*, with no wall-function or Sherwood-number closure. The rotating inner wall imposes solid-body azimuthal velocity $u_\theta = \omega R_i$; the Reynolds number $Re = \omega R_i d/\nu$ ranges from ≈ 160 at 100 rpm to ≈ 790 at the strongest bursts used by the RL agent (500 rpm), and the flow is treated as laminar.

The performance metric is the outlet conversion

$$X = 1 - \frac{\bar{c}_{\mathrm{out}}}{c_0}, \tag{4}$$

where $\bar{c}_{\text{out}}$ is the flux-weighted (cup-mixing) mean concentration over the side outlet, sampled every 0.1 s.

2.2 Rotation waveforms and motor power

The inner-cylinder speed follows the square-pulse family of Ref. [4]: bursts at ω_{hi} for a fraction D (duty cycle) of each period T, and a trough speed ω_{low} for the remainder, with 0.05 s ramped edges. Given a nominal (time-mean) speed w_b, the burst speed is *solved* from the mean constraint

$$\omega_{hi} = \frac{w_b - (1 - D)\,\omega_{low}}{D}, \tag{5}$$

so every waveform in the family has the same commanded time-average speed. Motor electrical power P is evaluated on the commanded $\omega(t)$ with the motor model of Ref. [4] (their Eqs. 18–23), densified at 100 Hz; because of Eq. (5), all waveforms at the same w_b draw nearly identical power (≈ 3.7 W at $w_b = 300$ rpm), which makes every comparison in this work an *equal-power* comparison. Power is normalized by $P_{\max} = 31.94$ W, the motor power at 2500 rpm, the fastest burst ever commanded.

2.3 Reinforcement-learning formulation

The CFD solver is wrapped in a Gymnasium [5] environment. An episode is 50 s of reactor operation from a pristine pre-filled initial condition ($c = c_0$ everywhere, fluid at rest), divided into five 10 s control blocks. At each block boundary the agent picks a 3-dimensional action, decoded to waveform parameters held for that block: duty $D \in [0.6, 1]$ (linear map), trough speed $\omega_{low} \in [0, 300]$ rpm (linear map), and period $T \in [0.5, 5]$ s (logarithmic map). The nominal speed is fixed at $w_b = 300$ rpm and the burst follows Eq. (5), capping bursts at 500 rpm. The wave phase resets at each block boundary (every block opens with a burst), a structural prior that keeps the commanded waveform fully determined by the action.

The observation at each block boundary is 7-dimensional: the block-averaged catalytic wall flux (normalized; the fast film-state signal), the change in conversion over the last block, the block-averaged conversion itself, the episode clock $t/50$ s, and the previous raw action. All flow observables are block *averages*, never point samples, which suppresses aliasing of the ± 0.02 outlet flutter. The per-block reward is

$$R_{\text{block}} = X_{\text{block}} - \frac{P_{\text{block}}}{P_{\max}}, \tag{6}$$

whose final-block value for a static policy equals the benchmark score $R = X_{[40,50]} - P/P_{\max}$ used throughout, so learned and static policies are directly comparable.

The agent is TD3 [2] (two 400×300 ReLU critics, delayed actor, target smoothing; discount 0.99, soft-update $\tau = 0.005$, exploration noise 0.2 per raw action dimension). Because one environment step costs ∼8 CPU-minutes of CFD, training is parallelized: 44 workers on one 48-core HPC node, each running its own OpenFOAM case, feed a single shared replay buffer while one learner trains the policy (308 episodes = 1,540 transitions per seed; the first 200 episodes are uniform-random exploration). Two fixes proved essential at this data scale. First, episodes end by *truncation* at the fixed horizon, and the stored done-flag must terminate the value bootstrap there: bootstrapping into end-of-episode states that never occur as source states lets the critic drift and drags the actor into action-space corners (observed in four failed runs). Second, the default one gradient step per collected transition badly underfits the critic at ∼1,500 transitions; an update-to-data ratio of 64, selected on a surrogate model fitted to 6,600 real CFD transitions, raised the surrogate convergence rate from 0/10 seeds to 7/10. With both fixes, two of three real-CFD seeds converged; all results below use the best seed, evaluated deterministically (exploration noise off) on a fresh case.

3 Results

3.1 Static modulation beats constant rotation

Table 2 summarizes the pulsed-vs-constant sweeps (the recreation of Fig. 7 of Ref. [4]) with resolved $Sc = 1075$ film physics. On the short reactor the paper's effect is reproduced but modest: at $w_b = 300$ rpm the $D = 80\%$, $T = 10$ s pulse gains +0.034 conversion over constant rotation. A scan over the static (D, T) family found the best fixed waveform (the "static champion") at $D = 80\%$, $T = 2.5$ s: $X = 0.392$, $R = 0.275$.

Constructing a benchmark stronger than the paper's. A claim of "beating the benchmarks" is only as strong as the benchmark, so we state precisely what the RL agent is compared against and why. All policies — constant, pulsed, and learned — are scored on the same conversion-vs-motor-power plane: the same solver and mesh, the same conversion metric (flux-weighted outlet mean over the last full period), and the same motor model of Ref. [4] evaluated on the commanded $\omega(t)$. On this plane "A beats B" has an unambiguous meaning: more conversion at equal or lower power.

We first recreated the paper's own headline waveform — $D = 20\%$, $T = 25$ s, i.e. 5 s bursts at $5w_b$ — on the short reactor. It *loses* to constant rotation at every nominal speed and every power (the lowest curve in Fig. 1): with $\tau/T \approx 1$, a fluid parcel transits the reactor within about one pulse period, so the 20 s idle phases starve the outlet faster than the bursts can compensate. This is a geometry effect, not a contradiction of the paper — the same waveform family wins handily in the tall reactor's amortization regime (Fig. 3) — but it means the literal paper configuration would be a strawman benchmark on the training geometry. We therefore built a stronger one: a grid over the whole waveform family, duty 15–80%, period 1.25–25 s, and nominal speed 100–500 rpm (Fig. 1). The grid shows that duty — not burst height — is the dominant knob: the high-duty, short-idle families (D = 70–80%, T = 10 s) dominate the low-power frontier, while the paper's deep 20% duty is the worst family tested. Refining the period at the best duty (Fig. 2) reveals a payoff plateau for $T \leq 2.5$ s: at $D = 80\%$ the idle phase must stay shorter than the $\sim$0.6 s vortex-relaxation time of the gap, or the film regrows between bursts. The frontier point at $w_b = 300$ rpm — $D = 80\%$, $T = 2.5$ s, $X = 0.392$, $R = 0.275$ — is the *static champion* against which all RL results are held. Beating it is a strictly harder test than beating the published waveform.

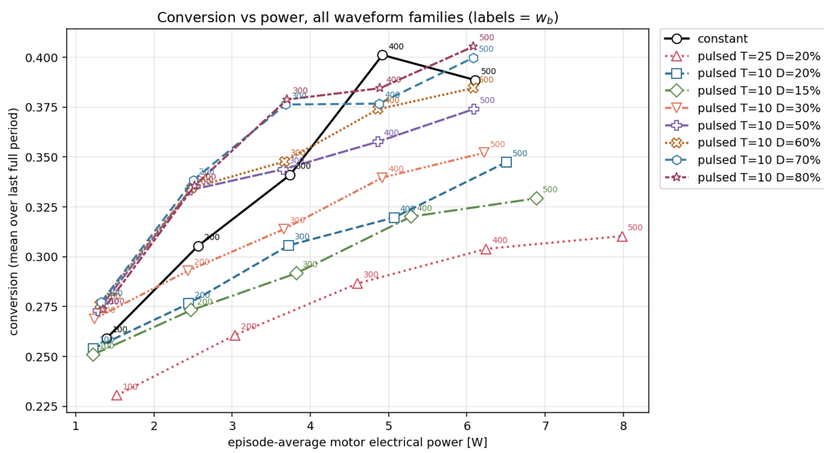

Figure 1: The static benchmark atlas (short reactor): conversion vs. episode-average motor power for constant rotation and eight pulsed waveform families (point labels = nominal speed w_b in rpm). The paper's original configuration ($T = 25$ s, $D = 20\%$, dotted) is the weakest family on this geometry; high-duty, short-period families (D = 70–80%, T = 10 s) form the low-power frontier.

On the full-height reactor (Fig. 3) the advantage is much larger and universal: pulsation beats constant rotation at *every* nominal speed, by up to +0.092 conversion, and pulsed conversion saturates at ≈ 0.877 from $w_b \geq 300$ rpm while constant rotation is still climbing at 500 rpm. The mechanism is the amortization regime identified in the paper: with $\tau/T \approx 13$, each fluid parcel averages many pulse periods, so idle phases cannot starve the outlet while the bursts repeatedly renew the wall film. Notably, pulsed $w_b = 200$ rpm (2.52 W) out-converts constant 500 rpm (6.11 W) — more conversion for 2.4× less power.

3.2 TD3 learns a state-dependent schedule that beats the static champion

Figure 4 shows the training curves of the three seeds. The best seed sustains a final-block reward above the pre-registered discovery threshold of $R = 0.28$ *with*

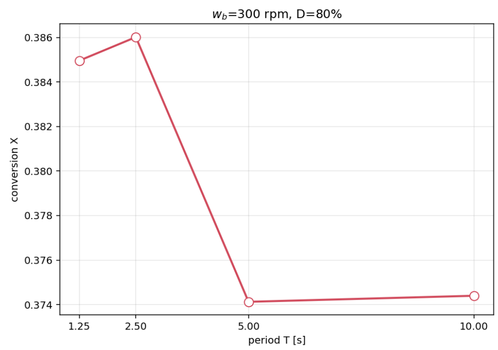

Figure 2: Period refinement at the best duty (w_b = 300 rpm, D = 80%): conversion is flat for $T \leq 2.5$ s and drops sharply once the idle phase ($0.2\,T$) exceeds the ∼0.6 s vortex-relaxation time. $T = 2.5$ s is the static champion.

Table 2: Pulsed (D = 80%, T = 10 s) vs. constant rotation at equal nominal speed. X = mean outlet conversion over the last full period of the episode; $R = X - P/P_{\max}$. Short reactor: 50 s episodes; full-height: 300 s episodes.

	Short ($\Gamma = 6$)			Full ($\Gamma = 30$)		
w_b [rpm]	X_{const}	X_{pulsed}	ΔX	X_{const}	X_{pulsed}	ΔX
100	0.258	0.272	+0.013	0.689	0.711	+0.022
200	0.307	0.333	+0.026	0.762	0.853	+0.092
300	0.340	0.374	+0.034	0.787	0.876	+0.090
400	0.399	0.374	−0.024	0.812	0.874	+0.063
500	0.392	0.396	+0.004	0.828	0.877	+0.049

exploration noise still on; a second seed converges to the static-champion level; the third falls into a known shallow trap (near-constant rotation) and only partially recovers. No seed exhibits the corner-pinning failure of the unfixed learner.

Evaluated deterministically, the learned policy earns $R = 0.289$ — $X = 0.406$ at 3.71 W — beating the static champion by +0.014 conversion at equal power, outside the measured ±0.01 chaos noise floor of the environment (Table 3, Fig. 6). Figure 5 places this result on the benchmark atlas of Fig. 1: the learned policy sits *above the entire static cloud* — every duty, every period, every nominal speed tested, including the champion at the same abscissa — at a power below that of any static policy of comparable conversion (constant 400 rpm reaches $X = 0.399$ but costs 4.92 W, 33% more). Because the fixed-mean constraint pins the commanded mean at 300 rpm, this is not a power-for-conversion trade: the gain comes entirely from *when* the modulation is applied. This is the sense in which the RL agent beats the benchmarks of Ref. [4]: not

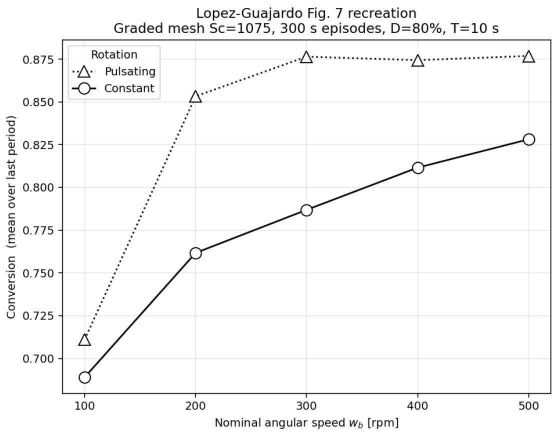

Figure 3: Recreation of Fig. 7 of Ref. [4] on the full-height ($\Gamma = 30$) reactor with resolved $Sc = 1075$ film physics: conversion vs. nominal speed for pulsed ($D = 80\%$, $T = 10$ s) and constant rotation. Pulsation wins at every speed and saturates the reactor's practical ceiling at roughly half the nominal speed.

merely their published waveform (which the static grid already dominates on this geometry), but the best member of their entire waveform family, re-tuned specifically for this reactor. The strategy is interpretable and genuinely state-dependent rather than a single waveform: (i) a gentle spin-up block ($D = 0.92$, trough 31 rpm, $T = 4.4$ s); (ii–iii) a hard, fast "shimmy" ($D \approx 0.6$, trough 0, $T = 0.5$–0.8 s, bursts to ≈ 490 rpm) while conversion builds; (iv) one slow, deep renewal pulse ($T = 5$ s); and (v) a final *constant*-300 "harvest" block ($D = 1$).

Table 3: Deterministic evaluation on the short reactor (50 s episodes, X over $[40, 50]$ s, all at commanded mean 300 rpm ≈ 3.7 W).

policy	X	P [W]	$R = X - P/P_{\max}$
constant 300 rpm	0.340	3.74	0.223
pulsed D=80%, T=10 s	0.374	3.70	0.259
pulsed D=80%, T=2.5 s (static champion)	0.392	3.72	0.275
TD3 (best seed, deterministic)	**0.406**	3.71	**0.289**

3.3 Zero-shot transfer to the full-height reactor

The trained policy was then deployed, with no retraining, on the full-height ($\Gamma = 30$) reactor in 300 s episodes (30 blocks). The only ambiguous input is the episode-clock observation; we tested the two natural deployments: *stretch* (feed the native clock $t/300$: the learned 50 s program dilated 6×) and *wrap*

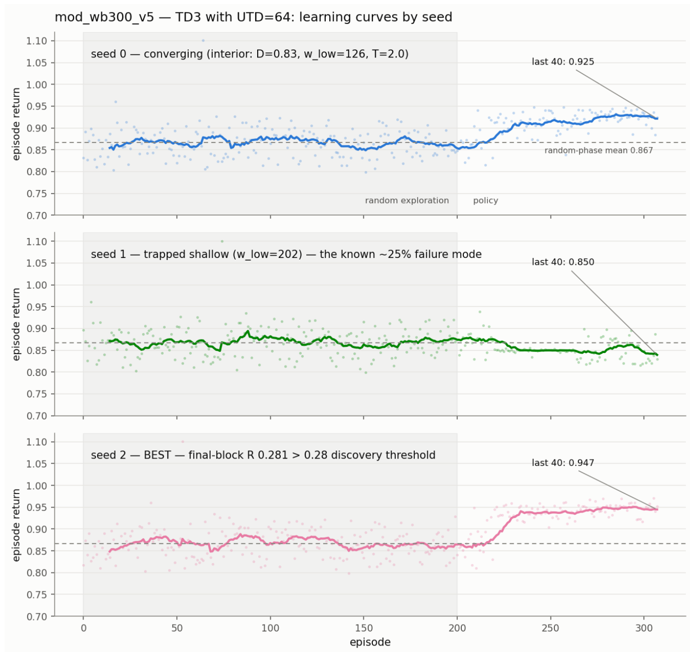

Figure 4: TD3 training on the short reactor (three seeds; 44 parallel CFD workers per seed, 308 episodes each). Horizontal lines mark the constant-300 baseline, the static champion ($R = 0.270$–0.275), and the $R = 0.28$ discovery threshold.

(feed $(t \bmod 50)/50$: the program replayed cyclically). Table 4 and Fig. 7 give the outcome: both variants beat constant rotation decisively, and the stretch deployment beats *every* static waveform measured on this reactor, at equal power.

Two observations argue that the policy is using state feedback rather than replaying a memorized script. First, in the wrap deployment the clock repeatedly visits the value at which, in training, the policy always switched to its harvest block — yet with conversion still climbing it declines to harvest and keeps shimmying. Second, in the stretch deployment the switch from shimmy to slow renewal and finally to harvest happens over blocks 19–30, re-timed to the tall reactor's much slower conversion transient, and the duty cycle rises *continuously* toward 1 rather than jumping at its trained schedule position.

4 Discussion

Mechanism. All gains are consistent with the boundary-layer renewal picture of Ref. [4]: at $Sc \sim 10^3$ the concentration film at the catalytic wall is thin and easily starved; bursts re-thin it and stir fresh reactant to the wall, while the

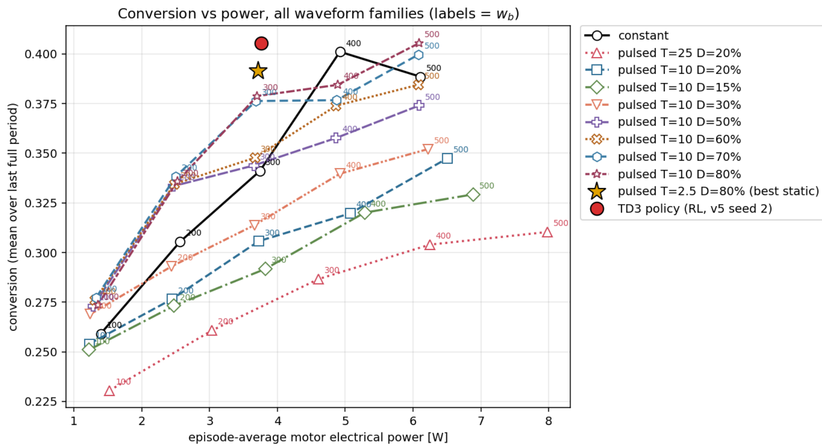

Figure 5: The head-to-head comparison: the deterministic TD3 policy (red dot) on the static benchmark atlas of Fig. 1. At equal power (3.71 W) it clears the static champion (gold star, $D = 80\%$, $T = 2.5$ s) by +0.014 conversion — beyond the ±0.01 noise floor — and sits above every static waveform family tested.

fixed-mean constraint means this costs no extra motor power. The learned policy improves on static pulsing mainly by *phasing*: aggressive renewal while conversion is building, gentler deep pulses once the film approaches its renewed state, and no wasted modulation once further renewal cannot pay back before the horizon.

The harvest is a finite-horizon effect. The terminal constant-speed block deserves scrutiny. Training optimizes a fixed 50 s episode from a known start — effectively a *batch campaign* objective, with the episode clock observable — so the rational endgame is to stop investing in future film state and cash out the flow state already built. The transfer runs expose this cleanly: constant-300 rotation *sustains* only $X = 0.787$ on the tall reactor, so the $X = 0.902$ read while rotating near-constant in the final blocks is necessarily transient — spent film capital, collectible once per campaign. The wrap deployment, whose clock never approaches the horizon, never harvests and *sustains* $X \approx 0.88$; that ($\approx$ tied with the best static pulse, +0.09 over constant) is the honest steady-state value of the learned controller, while the additional $\approx +0.02$ is real but end-of-campaign-only. For batch-style operation the policy is correct as-is; optimizing continuous operation instead calls for a continuing-task formulation — clock removed from the observation, soft resets or randomized long horizons, and bootstrapping at truncation — or, without retraining, deploying with the clock frozen mid-program, which reproduces the sustained shimmy-renewal regime.

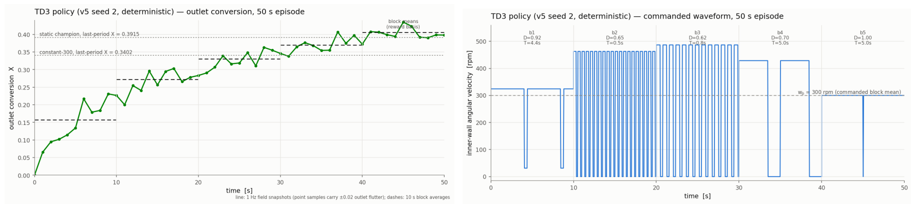

Figure 6: Deterministic evaluation of the learned policy on the short reactor. Left: outlet conversion vs. time against the static references. Right: the commanded speed — spin-up, fast shimmy, deep renewal, and constant harvest.

Table 4: Transfer to the full-height reactor (300 s episodes, X over the final 10 s window, equal power $\approx$ 3.7–3.8 W). References are the static sweep of Fig. 3 at $w_b = 300$ rpm.

policy	X	P [W]	R
constant 300 rpm	0.787	3.74	0.670
pulsed D=80%, T=10 s (best static @300)	0.876	3.70	0.760
TD3, wrap clock	0.881	3.78	0.763
TD3, stretch clock	**0.902**	3.78	**0.783**

Limitations. The domain is a 5° wedge, which suppresses non-axisymmetric azimuthal modes; the flow is assumed laminar throughout, reasonable at the operating $Re \lesssim 800$ but untested at the sweep's strongest bursts; episode metrics in the wavy regime carry a ± 0.01 reproducibility noise floor, so margins below that are not resolvable (the main claims clear it). The headline results come from one converged seed (one of three), evaluated deterministically; on the tall reactor the wall-flux observation exceeds its training normalization range by $\sim$60%, which the policy tolerated but which was not designed for. Motor power is computed from the commanded waveform via the motor model of Ref. [4], not from a coupled electromechanical simulation.

Future work. Freeing the mean-speed constraint (the environment already supports a free-mean action space) would let the agent trade power against conversion directly; replicate evaluations and the remaining seeds would tighten the statistics; and a continuing-task retraining would target steady-state operation. On the experimental side, the policy's commanded waveforms are ordinary tabulated speed profiles, directly playable on the physical reactor of Ref. [4].

5 Conclusions

A TD3 agent, trained on a resolved-film CFD model of a catalytic-wall Taylor-Couette reactor under an equal-power constraint, discovered a state-dependent

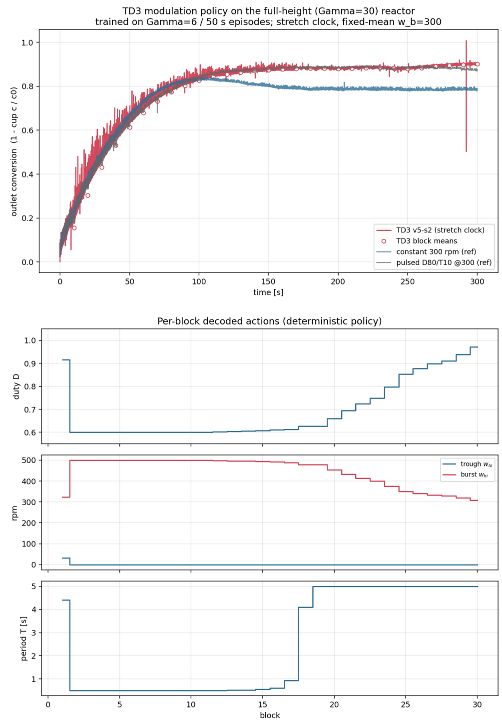

Figure 7: Zero-shot transfer (stretch clock). Top: conversion vs. time against the full-height static references. Bottom: the per-block decoded actions — the learned program dilates gracefully: shimmy through mid-episode, then a continuous glide into harvest (duty 0.63 → 0.97) as the clock approaches 1.

rotation-modulation schedule that beats the best static waveform on its training geometry ($R = 0.289$ vs. 0.275) and, deployed zero-shot, on the full-height reactor ($X = 0.902$ vs. 0.876 for the best static pulse and 0.787 for constant rotation, all at ≈ 3.7 W). The learned strategy is interpretable — spin-up, fast film-renewal shimmy, deep slow pulses, terminal harvest — and its advantage decomposes into a sustainable feedback component and a finite-horizon harvest component, a distinction that matters whenever episodic RL is applied to continuously operated process equipment.

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